

People Reject Algorithms in Uncertain Decision Domains Because They Have Diminishing Sensitivity to Forecasting Error



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Abstract

Will people use self-driving cars, virtual doctors, and other algorithmic decision-makers if they outperform humans? The answer depends on the uncertainty inherent in the decision domain. We propose that people have diminishing sensitivity to forecasting error and that this preference results in people favoring riskier (and often worse-performing) decision-making methods, such as human judgment, in inherently uncertain domains. In nine studies ($N = 4,820$), we found that (a) people have diminishing sensitivity to each marginal unit of error that a forecast produces, (b) people are less likely to use the best possible algorithm in decision domains that are more unpredictable, (c) people choose between decision-making methods on the basis of the perceived likelihood of those methods producing a near-perfect answer, and (d) people prefer methods that exhibit higher variance in performance (all else being equal). To the extent that investing, medical decision-making, and other domains are inherently uncertain, people may be unwilling to use even the best possible algorithm in those domains.

Keywords

judgment, decision-making, algorithm aversion, forecasting, variance, open data, open materials, preregistered

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Will people use algorithmic decision-makers—such as self-driving cars, robot doctors, and virtual hiring and admissions officers—once the technology is widely available? How well will these algorithms need to perform for people to embrace them?

In many decision domains, algorithms¹ already outperform humans. Calculators and analytical tools make fewer mathematical errors than human problem solvers, and in more uncertain domains, algorithmic forecasters outperform human forecasters on average (Dawes, Faust, & Meehl, 1989; Meehl, 1954). Meta-analyses investigating predictions in the domains of clinical health (Ægisdóttir et al., 2006); human health and behavior (Grove, Zald, Lebow, Snitz, & Nelson, 2000); medicine, business, psychology, and education (Kaufmann & Wittmann, 2016); and hiring and academic admissions (Kuncel, Klieger, Connelly, & Ones, 2013) have found that algorithms consistently outperform human judgment. Even simplistic linear models beat experts (Dawes, 1979), and models

designed to distill an expert's prediction process almost always outperform the expert they were based on (Camerer, 1981).

Yet even when algorithms are the best forecasting options available, people often underuse them. Firms (Sanders & Manrodt, 2003) and professionals (Fildes & Goodwin, 2007; Vrieze & Grove, 2009) regularly choose not to use algorithms as their primary method of forecasting, and this often results in less accurate forecasts (Sanders & Manrodt, 2003). Additionally, participants in experimental judgment tasks often reject simple decision rules that outperform human judgment (Arkes, Dawes, & Christensen, 1986; Estes & Straughan, 1954). Participants are also less likely to rely on algorithmic

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forecasts after seeing them err, even when they believe that the algorithm outperforms the human alternative on average (Dietvorst, Simmons, & Massey, 2015). This unwillingness to adopt algorithms that outperform humans can have an enormous cost; for example, the majority of Americans report that they would not be comfortable riding in a self-driving car (Leinert, 2018), but research suggests that early adoption of self-driving cars could save hundreds of thousands of lives (Kalra & Groves, 2017).

Although the existing literature supports the notion that people often underuse algorithms that outperform humans on average, it does not explain why people underuse these algorithms. We propose that people have diminishing sensitivity to forecasting error, which causes them to prefer decision-making methods that they believe have the highest likelihood of providing a near-perfect answer (i.e., one with little to no error). This decision strategy encourages risk-taking and results in people favoring riskier, and often worse-performing, decision-making methods (such as human judgment) when they feel that an algorithm is unlikely to generate a near-perfect answer. Further, as explained in the next section, this strategy causes people to reject even the best possible algorithm in inherently uncertain domains. The implications are substantial; to the extent that medical decision-making, consumer-demand forecasting, and other domains are inherently uncertain, people may be unwilling to use even the best possible algorithms in those domains.

People Have Diminishing Sensitivity to Forecasting Error

Performance in a prediction² task can be quantified as error—that is, the difference between the response and the outcome. When decision-makers choose between different forecasting methods on the basis of performance, they consult their beliefs about each method's likelihood of making errors of different magnitudes. All decision-makers would prefer a forecasting method that makes no errors to one that always makes large errors; however, their preferences when trading off among small, medium, and large errors depend on how sensitive they are to errors of different magnitudes.

We propose that people have diminishing sensitivity to forecasting error. Decision-makers who have diminishing sensitivity to error face a smaller penalty for each marginal unit of error that a forecast produces (see Fig. 1). As a result, they feel relatively large subjective differences between different magnitudes of near-perfect forecasts (the best possible forecasts that produce little to no error) and relatively small subjective differences between forecasts with greater amounts of error.

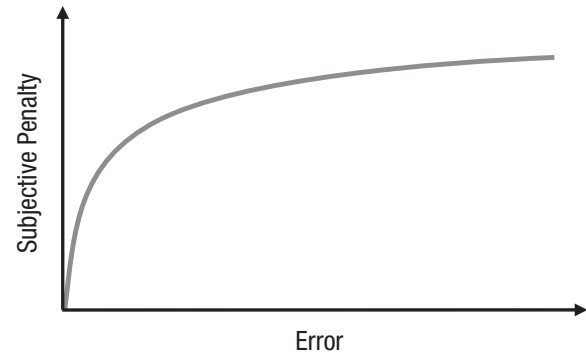


Fig. 1. Penalty function showing the subjective penalty that a decision-maker with diminishing sensitivity to error could face (y -axis), given the error generated by a forecast (x -axis).

Diminishing sensitivity to forecasting error has important implications for people's selection of forecasting methods. People who have diminishing sensitivity to error prefer methods that exhibit higher variability in performance while holding average performance fixed. In other words, these people are risk-seeking. Higher variance methods have the advantage of being more likely to produce near-perfect answers and the corresponding disadvantage of being more likely to produce very inaccurate answers; choosers with diminishing sensitivity give greater weight to the upside. Relatedly, when decision-makers with diminishing sensitivity to forecasting error compare the performance of two different forecasting methods, they will give disproportionate weight to their expectations about near-perfect forecasts. As a result, they will not always favor the forecasting method that offers better average performance—they may favor a method that they expect to provide more near-perfect forecasts instead. These preferences are important to consider in the context of human and algorithmic forecasters.

Choosers With Diminishing Sensitivity to Error May Not Prefer Algorithms

Algorithms typically offer lower variance than human judgment. This is, at least in part, because algorithmic decision-makers are more consistent: Given the same inputs, algorithms always generate the same answer, but the same cannot be said for humans (see Dawes et al., 1989). Further, an algorithm's general approach to forecasting is constant, whereas human judges can change their approach at will. Thus, the future performance of human forecasters is often more variable and uncertain in expectation.

Given the typical characteristics of human and algorithmic forecasters, a choice between the two often consists of (a) an option with higher expected average

performance and lower expected variance of performance (the algorithm) and (b) an option with lower expected average performance and higher expected variance of performance (the human). Under these conditions, decision-makers with diminishing sensitivity to error could justifiably choose either option, depending on their specific expectations and subjective penalty function. However, we posit that their choices may be particularly sensitive to the uncertainty inherent in the decision domain.

Diminishing Sensitivity Leads to Underuse of Algorithms in Uncertain Domains

We propose that people will be especially likely to prefer human forecasters to algorithmic forecasters in decision domains with more irreducible uncertainty. Irreducible uncertainty is uncertainty that cannot be resolved until the event in question takes place; for example, uncertainty about the outcome of a die roll is irreducible because it cannot be resolved until the die is rolled. Real-world prediction domains, such as stock prices or consumer preferences, contain irreducible uncertainty to the extent that outcomes in those domains are inherently unpredictable.

In deterministic domains, in which there is no irreducible uncertainty (e.g., solving math problems, recalling established facts), the best possible decision-making method can always produce an answer with no error. For example, a calculator can always provide the exact answer to a multiplication problem. However, in decision domains with irreducible uncertainty, even the best possible decision-making method will make errors because outcomes are inherently unpredictable. For example, even the best possible strategy for playing blackjack results in about as many hands lost as won (Fogel, 2004) because of uncertainty about hidden cards.

As irreducible uncertainty in a decision domain increases, algorithmic forecasting methods may become relatively less attractive to choosers with diminishing sensitivity. The performance of all forecasting methods tends to degrade and converge as irreducible uncertainty increases—random noise makes the true “skill” of each forecaster harder to detect. Additionally, as the irreducible uncertainty in a domain increases, all forecasters’ likelihood of making a near-perfect forecast typically decreases because making a perfect or near-perfect forecast increasingly becomes a matter of luck. In other words, in domains with greater uncertainty, the gap between human and algorithmic performance is typically smaller, and both humans and algorithms provide fewer near-perfect forecasts. Under these conditions, the higher variance in performance expected from human forecasters is more likely to sway choosers with diminishing sensitivity to error.

To summarize, in deterministic domains, it will be clear that well-designed algorithms are more likely than humans to provide near-perfect answers, and, accordingly, choosers with diminishing sensitivity to error will tend to pick those algorithms. However, as irreducible uncertainty in the decision domain increases, the performance differences between human and algorithmic forecasters will diminish, and both will be relatively unlikely to generate near-perfect forecasts. As a result, choosers with diminishing sensitivity to forecasting error may prefer to rely on riskier, higher-variance human judgment.

The Present Studies

Our propositions led us to a number of predictions that we tested empirically. In Study 1, we found evidence that people have diminishing sensitivity to forecasting error. In Study 2, we found that people prefer forecasting methods that exhibit higher variance in performance, consistent with diminishing sensitivity to error. In Studies 3 through 5, we investigated the implications of diminishing sensitivity to error for choices between algorithmic and human forecasting methods. We found that people are more likely to reject algorithms in more uncertain domains (Study 3), that their preference for human forecasters in uncertain domains is based on expectations about near-perfect forecasts (Study 4), and that they are more likely to choose algorithms that exhibit higher variance in performance (Studies 5a and 5b).

In Studies 2 through 5, we created judgment tasks in which two different decision methods predicted an outcome number that was a function of three provided input numbers. This setup mimics real-life scenarios, such as assessing job applicants, products, or investments, in which decision-makers generate forecasts on the basis of a set of attributes. Participants completed 20 practice trials in which each decision method made a forecast, and participants received feedback on each method’s performance after each trial. After these 20 trials, participants decided which method they would like to use for an incentivized trial in which they would gain more money the closer the forecast of their chosen method was to the actual outcome. Finally, participants reported their confidence in each method and their beliefs about each method’s performance.

In Studies 2 through 5, we manipulated the performance of the algorithm by altering the irreducible uncertainty in the task between conditions. This manipulation allowed us to alter aspects of the algorithm’s performance while holding the algorithm itself constant. The algorithm (identified as “the model” to participants) was always the optimal decision-making method for the judgment task. The model used the same equation that generated the outcome omitting any irreducible uncertainty. As a result, the model made the

Table 1. Distributions of Model Performance in Study 1

Performance	Model A	Model B: upgrade-to- perfect condition	Model B: upgrade-to-1 condition	Model B: upgrade-to-2 condition
Perfect (\$1)	20%	40%	20%	20%
Off by 1 (\$0.80)	20%	0%	40%	20%
Off by 2 (\$0.60)	20%	20%	0%	40%
Off by 3 (\$0.40)	20%	20%	20%	0%
Off by 4 (\$0.20)	20%	20%	20%	20%

Note: In the performance column, values in parentheses indicate the bonus amount participants received.

best possible forecasts, maximized expected earnings for participants, and could not be beaten in the long run. This design let us avoid the concern that participants who chose not to use the model would have used a better model, because no better model was possible.

For each study, we set the sample size before any data were obtained, and we collected 200 observations per condition. We report all exclusions (if any), all manipulations, and all measures in the article or in the supplemental materials available on OSF (<https://osf.io/erjq9/>). For all studies, preregistrations, raw data, analysis code, original materials, and detailed descriptions, as well as additional studies that conceptually replicated our findings, have been posted on OSF (<https://osf.io/erjq9/>). We focus on the key elements of each study here and describe the studies and results in full detail in Sections S1 through S5 in the supplemental materials.

Study 1

In Study 1, we tested the proposition that people have diminishing sensitivity to forecasting error. Participants learned that they would choose between two models (model A and model B) for a forecasting task. Model B was a strict upgrade over model A. We manipulated where in the distribution of performance model B had an advantage and asked participants to report their willingness to pay to use model B. We expected that diminishing sensitivity to error would lead participants to demonstrate higher willingness to pay for model B when its advantage over model A was in performances closer to perfection.

Method

We preregistered that we would recruit 600 participants from Amazon Mechanical Turk (200 per condition). After the study was posted, 678 participants clicked on the study link; 40 participants failed an attention check

and were not allowed to begin the study, and 601 participants completed all dependent measures. This sample was 57% male and 36 years old on average.

Participants who passed the attention check learned that their job was to learn about the performance of two different models in a forecasting task and then make a decision that would determine which model determined their bonus. Participants read that both models predicted a numerical outcome from 0 to 10. Participants read that they would earn a \$1 bonus if the forecast were perfect and that their bonus would decrease by \$0.20 for each unit of error in the forecast (see Section S7 of the Supplemental Materials document at <https://osf.io/erjq9/> for a description of each study's bonus scheme).

Next, participants learned about the two models, starting with model A. They read that model A was equally likely to be perfect, off by 1, off by 2, off by 3, and off by 4, and they saw a chart that showed the probability of each outcome (20%) and the frequency of each outcome (e.g., model A is perfect in one out of every five trials on average). We explicitly told participants each model's distribution of performance so they would not have to rely on expectations. Given the bonus scheme, model A earned \$0.60 on average. On the next screen, participants learned about model B.

Each participant read about one of three different versions of model B, depending on their between-subjects condition (see Table 1). Model B was similar to model A, but all of the performances that were off by a certain amount (e.g., off by 2) were upgraded by 1 unit of error (e.g., to off by 1). Participants in the *upgrade-to-perfect* condition saw a version of model B that 40% of the time was perfect; that was never off by 1; and that 20% of the time was off by 2, 3, and 4, respectively. Participants in the *upgrade-to-1* condition saw a version of model B that 40% of the time was off by 1; that was never off by 2; and that 20% of the time was perfect, off by 3, and off by 4, respectively. Participants in the *upgrade-to-2* condition saw a version of

model B that 40% of the time was off by 2; that was never off by 3; and that 20% of the time was perfect, off by 1, and off by 4, respectively. We held the set probabilities associated with the different potential performances of model B constant (0%, 20%, and 40%) to make the design more robust to various probability-weighting functions (Kahneman & Tversky, 1979). Participants read text describing the differences between model A and model B and saw a chart that showed the probability and frequency of each outcome for model B.

All versions of model B were strict upgrades over model A and generated bonuses with the same expected value (\$0.64); however, the different versions of model B produced errors of different magnitudes with different frequencies. Thus, participants' willingness to pay for the different versions of model B demonstrated their sensitivity to errors of different magnitudes. A participant with a constant sensitivity to error would value each version of model B equally because each offered an average error of 1.8. A participant with increasing sensitivity to error (or decreasing sensitivity to gains, as in prospect theory; Kahneman & Tversky, 1979) would value the upgrade-to-2 version of the model over the upgrade-to-1 version and would value the latter over the upgrade-to-perfect version because they would feel a larger subjective difference between relatively large errors such as 3 and 2 (and small gains such as \$0.40 and \$0.60) than between relatively small errors such as 1 and 0 (and large gains such as \$0.80 and \$1.00). On the other hand, a participant with decreasing sensitivity to error would value the upgrade-to-perfect version of the model over the upgrade-to-1 version and would value the latter over the upgrade-to-2 version because they would feel a larger subjective difference between relatively small errors such as 1 and 0 than relatively large errors such as 3 and 2. In other words, only participants with decreasing sensitivity to error would place a greater value on a 1-point improvement in error when it is applied to performances closer to perfection.

Next, participants learned about the process that would determine which model they would use. Participants read that they would have to pay to use model B if they would like to use it instead of model A and that they would bid from \$0.00 to \$0.10 to report their willingness to pay for model B. Participants learned that the computer would randomly select a value from 1 to 10 cents to be the price of model B. If their bid were more than or equal to the price, they would pay the price (deducted from their bonus) and use model B. If their bid were less than the price, they would not pay the price and use model A.

On the next screen, participants made their bid for model B. The differences between model A and model B were restated on screen, along with a chart that

depicted the probability with which model A and model B realized errors of each possible magnitude. At the bottom of the screen, participants completed their bid on an 11-point Likert scale from \$0.00 to \$0.10. Finally, participants reported their age, sex, and highest level of education and completed the study.

Results

Consistent with diminishing sensitivity to forecasting error, results showed that participants were willing to pay more for model B when its performance advantage over model A was in performances closer to perfection (see Fig. 2). An ordinary least squares regression of willingness to pay on a condition variable coded 0, 1, or 2 (representing the upgrade-to-perfect, upgrade-to-1, and upgrade-to-2 conditions, respectively) revealed that participants had lower willingness to pay for model B the further its performance upgrade over model A was from perfection, $b = -0.58$, $t(599) = -3.86$, $p < .001$, $\eta^2 = .024$. Follow-up analyses comparing each pairing of conditions revealed that participants in the upgrade-to-perfect condition had higher willingness to pay than participants in the upgrade-to-1 condition, $t(401) = 1.81$, $p = .070$, $d = 0.18$, and the upgrade-to-2 condition, $t(398) = 3.80$, $p < .001$, $d = 0.38$, and that participants in the upgrade-to-1 condition had higher willingness to pay than participants in the upgrade-to-2 condition, $t(397) = 2.07$, $p = .039$, $d = 0.21$. Participants' lower willingness to pay for equally valuable upgrades that were further from perfection provides direct evidence that they had decreasing sensitivity to each marginal unit of forecasting error that they faced.

Study 2

Study 1 provided evidence that people have diminishing sensitivity to forecasting error. In Study 2, we tested an implication of this penalty function; people prefer forecasting methods that exhibit higher variance in performance, with average performance held constant.

Method

We preregistered that we would recruit 400 participants from Amazon Mechanical Turk (200 per condition). After we posted the study, 514 participants clicked on the study link; 81 participants failed an attention check and were not allowed to begin the study, and 403 participants completed all dependent measures. This sample was 46% male and 36 years old on average.

Participants who passed the attention check read about the simulated prediction task that we created, in which two models, model E and model S, predicted

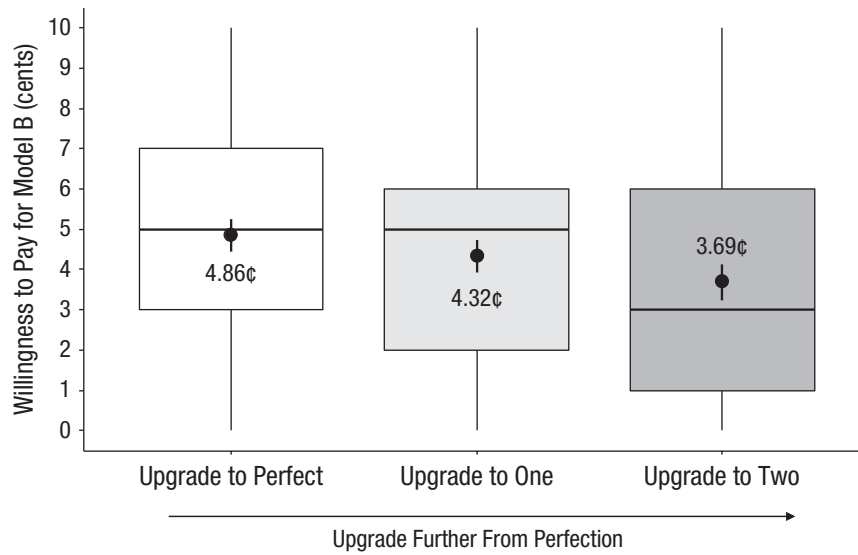


Fig. 2. Results for Study 1: mean amount that participants were willing to pay for model B in the three upgrade conditions. The black dots represent means, and error bars represent 95% confidence intervals. Horizontal lines indicate medians, and the boxes indicate interquartile ranges.

two separate outcomes that ranged from 0 to 10 on the basis of three variables that also ranged from 0 to 10. Unbeknownst to participants, the outcomes were a linear combination of the three inputs plus a random component, and the model that predicted each outcome was the same linear combination of the three inputs that generated the outcome without the random component (see Section S8 of the Supplemental Materials document at <https://osf.io/erjq9/> for a detailed description of all prediction tasks).

Participants read that the task would have two stages: the trials and the bonus stage. In each of the 20 trials, participants would see each model's forecast and the correct answers for one set of stimuli so they could judge the performance of each model. Then, during the bonus stage, they would decide which model they would like to rely on to generate a forecast that would determine their bonus.

Participants were assigned to two conditions. We manipulated the variance of model S's performance between conditions while holding average performance and earnings fixed for both models. We added random error so that model S was always off by 1 unit in the *low-variance* condition and was equally likely to be off by 0 or 2 units in the *high-variance* condition. Model E was always off by 1 unit in both conditions. Thus, both models were off by 1 unit on average, but in the high-variance condition, model S offered higher variance in performance. We expected participants to be more likely to choose model S in the high-variance

condition than the low-variance condition if people indeed have diminishing sensitivity to forecasting error.

After completing the 20 trials, participants learned that they could earn up to \$0.40 depending on the accuracy of their chosen model's prediction in the bonus round. They would earn \$0.40 if the model's prediction were perfect, and this bonus would decrease by \$0.04 for each unit of error in the model's forecast. Next, participants made a binary choice between using model E and model S to make a forecast that would generate their bonus. This choice between forecasting methods was the key dependent measure in each of Studies 2 through 5. Participants did not see the correct answer or the model's answer after making their choice.

Participants responded to a set of questions to complete the study. Participants reported an estimate of each model's average absolute error using text boxes and reported their confidence in each model's answers using 5-point scales. Finally, participants reported their age, sex, and level of education.

Results

Each model offered the same average performance by design, and participants reported that the average performance of the models was similar in both the high-variance condition, $t(203) = 1.18$, $p = .240$, $d = 0.06$, and the low-variance condition, $t(198) = 1.62$, $p = .108$, $d = 0.07$. However, although participants did not express significantly different levels of confidence in each

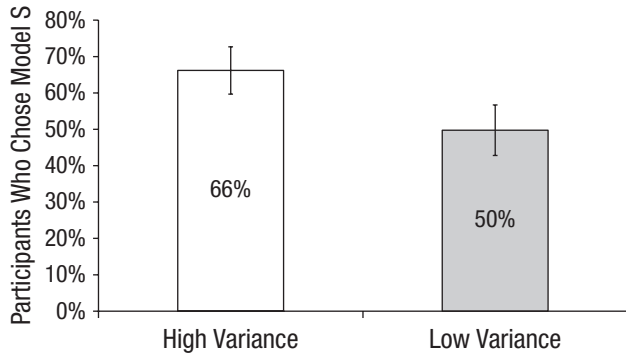


Fig. 3. Results for Study 2: percentage of participants who chose model S in both the high-variance condition (model S off by 0 or 2, model E off by 1) and low-variance condition (both models off by 1). Error bars represent 95% confidence intervals.

model in the low-variance condition (in which both models offered the same performance), $t(198) = -0.33$, $p = .743$, $d = -0.03$, participants in the high-variance condition expressed more confidence in model S (which offered higher variance in performance), $t(203) = -6.18$, $p < .001$, $d = -0.56$. Similarly, participants were significantly more likely to choose model S in the high-variance condition (66%) than in the low-variance condition (50%; see Fig. 3), $\chi^2(1, N = 403) = 11.16$, $p = .001$, Cramer's $\phi = .17$, consistent with the notion that they prefer higher-variance forecasting methods, holding average performance fixed. This increased preference for model S in the high-variance condition persisted in a logistic regression controlling for participants' estimates of both models' average performance, $z(403) = 4.00$, $p < .001$, odds ratio (OR) = 2.47, 95% confidence interval (CI) = [1.59, 3.85].

These findings are striking because the literature suggests that, all else being equal, people typically want to minimize variance in performance when they stand to gain money. In fact, in the domain of gains, people would typically pay to reduce variance in their earnings (Kahneman & Tversky, 1979). Participants' preference for a higher variance algorithm, however, supports the notion that people prefer higher variance in the performance of forecasting methods because they have diminishing sensitivity to forecasting error. In the remaining studies, we tested the implications of diminishing sensitivity to error for choices between humans and algorithms.

Study 3

In Study 3, we tested the hypothesis that people use algorithms less in domains with more irreducible uncertainty. We assigned participants to five conditions with increasing amounts of irreducible uncertainty, and

participants decided between using their own answer or a model's answer for an incentivized trial.

Method

We preregistered that we would recruit 1,000 participants from Amazon Mechanical Turk (200 per condition). After we posted the study, 1,233 participants clicked on the study link; 125 participants failed an attention check and were not allowed to begin the study, and 1,005 participants completed all dependent measures. This sample was 49% male and 36 years old on average.

Participants who passed the attention check read that they would learn about a prediction task by making a series of trial predictions and then have the opportunity to earn a bonus depending on the accuracy of an incentivized prediction. Next, participants learned that they would use three variables that ranged from 1 to 25 to predict an outcome that typically ranges from 20 to 80 units (all whole numbers). Participants read that a statistical model given the same information that they would receive would also make these predictions. Participants learned that after making each prediction in the 20 trial stages, they would see (a) their prediction, (b) the model's prediction, and (c) the actual answer, so they could learn about the task.

Participants were assigned to five conditions, and we manipulated the amount of irreducible uncertainty in the prediction task among these conditions. In the *certain* condition, the outcome was determined by an equation that combined the three inputs given to participants with no random component. In the *uncertain-1* condition, we added a random draw from the discrete uniform distribution of whole numbers from -3 to 3 to the same equation to generate the outcome. Similarly, in the *uncertain-2* through *uncertain-4* conditions, this discrete uniform distribution of whole numbers ranged from -6 to 6 , -9 to 9 , and -12 to 12 , respectively. Thus, the amount of irreducible uncertainty in the prediction task increased across the uncertain conditions while the rest of the task was held constant (see Section S8 of the Supplemental Materials document for details).

In each condition, the model's prediction was the output of the equation that determined outcomes in the prediction task, excluding any random component. As a result, the model made the exact same predictions in each condition. In the *certain* condition, the model was always perfect, and in the uncertain conditions, its performance depended on the random draw. Because the model's prediction was always at the center of the random draw's uniform distribution, it always produced forecasts that maximized expected earnings and stochastically dominated all other forecasting strategies.

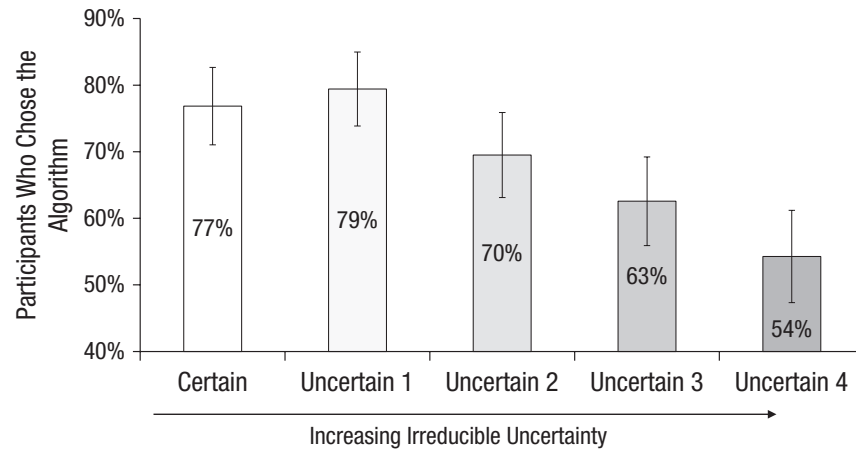


Fig. 4. Results for Study 3: percentage of participants who chose to use the algorithm's answer in each of the five conditions. Error bars represent 95% confidence intervals.

However, as the irreducible uncertainty in the task increased across the conditions, the model made fewer near-perfect forecasts. As a result, we expected participants to be less likely to use the model as the irreducible uncertainty in the task increased, even though it was the best possible forecasting method.

Next, participants made their 20 trial predictions. After completing these 20 trials, participants learned that they could earn up to \$0.50 depending on the accuracy of the incentivized prediction. They would earn \$0.50 if the incentivized prediction were off by up to 3 units, and this bonus would decrease by \$0.10 for each additional 3 units of error. Importantly, in this study (and all studies), all of the model's predictions were good enough to earn a bonus. The only way that participants could earn no bonus was by choosing not to use the model's estimate and subsequently making a poor estimate themselves.

Next, participants made a binary choice between using either their own answer or the model's answer as the incentivized prediction that would determine their bonus. Participants learned that they would generate their own answer no matter which option they chose; then they made their predictions. Participants did not see the correct answer or the model's answer.

Participants responded to a set of questions to complete the study. Participants entered an estimate of their own average absolute error and the model's average absolute error in text boxes, and they rated their confidence in the model's answers and their own answers using 5-point scales. Finally, participants reported their age, sex, and level of education.

Results

The model performed substantially better than participants during the practice trials—it provided answers

that were at least 130% better (i.e., 2.3 times better) than participants' answers on average in each condition, certain: $t(202) = 45.81, p < .001, d = 4.55$; uncertain 1: $t(203) = 55.35, p < .001, d = 5.46$; uncertain 2: $t(199) = 43.24, p < .001, d = 4.32$; uncertain 3: $t(202) = 37.86, p < .001, d = 3.76$; uncertain 4: $t(198) = 35.62, p < .001, d = 3.59$. In fact, not one of the 1,005 participants in the study provided forecasts that were more accurate than the model's on average. Further, participants reported that the model performed better than they did on average, certain: $t(202) = -8.29, p < .001, d = -0.47$; uncertain 1: $t(203) = -9.48, p < .001, d = -0.46$, uncertain 2: $t(199) = -10.82, p < .001, d = -0.39$; uncertain 3: $t(200) = -7.35, p < .001, d = -0.27$; uncertain 4: $t(197) = -5.06, p < .001, d = -0.22$, and reported more confidence in the model's answers than their own on average in each condition, certain: $t(202) = 14.74, p < .001, d = 1.48$; uncertain 1: $t(203) = 16.26, p < .001, d = 1.51$; uncertain 2: $t(199) = 13.92, p < .001, d = 1.28$; uncertain 3: $t(200) = 11.62, p < .001, d = 1.11$; uncertain 4: $t(197) = 9.21, p < .001, d = 0.81$. However, even though participants recognized that the model performed better than they did on average, they chose it significantly less often the more irreducible uncertainty there was in the forecasting task (see Fig. 4).

A logistic regression of participants' choice of the model on a condition variable (coded 1 through 5) revealed that participants were significantly less likely to choose to use the model's answer the more irreducible uncertainty there was in the task, $z(1009) = -5.91, p < .001, OR = 0.74, 95\% CI = [0.68, 0.82]$. Further, the negative effect of irreducible uncertainty persisted when participants' estimates of their own and the model's average performance were added to the logistic regression as controls, $z(1006) = -4.75, p < .001, OR = 0.78, 95\% CI = [0.70, 0.86]$, suggesting that participants' perceptions of average performance do not

account for the effect. The decision to reject the best possible algorithm was not limited to the minority of participants who did not recognize that it was the better option; more than half of the participants who chose not to use the model in the uncertain-4 condition estimated that the model performed better than they did on average. These results show that participants were less likely to use the algorithm as the irreducible uncertainty in the task increased and it made fewer near-perfect forecasts, even though the model was the best possible forecasting method.

In addition to the main findings, the results of Study 3 suggest that people do not abandon algorithms in reaction to seeing small amounts of error. Participants in the certain and uncertain-1 conditions picked the model at a similar frequency (77% and 79%), $\chi^2(1, N = 407) = 0.39, p = .531$, Cramer's $\phi = .03$, even though the model usually made errors in the uncertain-1 condition (in 86% of trials) and was always perfect in the certain condition. Thus, it seems unlikely that people's underuse of algorithms results from an absolute intolerance of any error, as past research has suggested (see Dietvorst et al., 2015; Einhorn, 1986; Highhouse, 2008).

Studies S1 through S3 (reported at <https://osf.io/erjq9/>) replicated the results of Study 3 under different conditions. We put these studies in the Supplemental Materials because they serve as conceptual replications of Study 3 that mainly rule out alternative explanations. Studies S1 and S2 demonstrate that participants placed in an uncertain condition recognize that the judgment task contains irreducible uncertainty and that participants who are explicitly informed about the irreducible nature of the uncertainty in the task exhibit similar behavior to those who are not. These results suggest that people's tendency to shift away from algorithms in uncertain domains is not driven by an expectation that absolutely perfect performance is possible in the long run. Study S3 conceptually replicated the negative effect of irreducible uncertainty on choice of a model using a judgment task built with real data, suggesting that the results were not due to the artificial nature of this task. Overall, the results of Study 3 and Studies S1 through S3 suggest that many people may be unwilling to use even the best possible algorithm in inherently uncertain domains.

Study 4

Study 4 built on Study 3 by investigating how people's decreased use of algorithms in uncertain domains related to their beliefs about future performance.

Method

We preregistered that we would recruit 400 participants from Amazon Mechanical Turk (200 per condition). After we posted the study, 554 participants clicked on the study link; 87 participants failed an attention check and were not allowed to begin the study, and 405 participants completed all dependent measures. This sample was 44% male and 38 years old on average.

Study 4 had a similar design to Study 3, but with a number of changes. We changed the prediction task to see whether the previous results would replicate under different conditions. Participants received three input numbers (ranging from 0 to 10) that were combined to produce the answer (ranging from 0 to 10). We assigned participants to two conditions. In the *certain* condition, no irreducible uncertainty was added to the task, and the model was always perfect. In the *uncertain* condition, we added a random draw from the discrete uniform distribution of whole numbers from -3 to 3 to the equation, and as a result, the model was always off by 0, 1, 2, or 3. Participants earned \$0.50 for a perfect forecast, and their bonus decreased by \$0.10 for each unit of error.

Additionally, we changed how we collected participants' beliefs about their own and the models' performance in order to gather mechanism evidence. Instead of asking for one estimate of average performance, we had participants report how they believed they and the model would perform in each of 20 future forecasts using the distribution builder tool (*distBuilder*; Version 1.1.0) created by André (2016). After the incentivized forecast, participants read a set of instructions about this tool and then completed their distribution for both the model's forecasts and their own forecasts, in randomized order (see Section S9 of the Supplemental Materials document at <https://osf.io/erjq9/> for full details).

If people have diminishing sensitivity to forecasting error, the difference in the frequency with which they expect their forecasting options to make a perfect forecast would be more influential than the difference between their expectations at any other point in the distribution of performance. Thus, we anticipated that participants' choice of forecasting method would relate to their expectations about which forecasting method was more likely to make a perfect forecast.

Results

As in Study 1, the model performed significantly better than participants on average, certain: $t(205) = 32.36, p < .001, d = 3.19$, uncertain: $t(203) = 18.62, p < .001$,

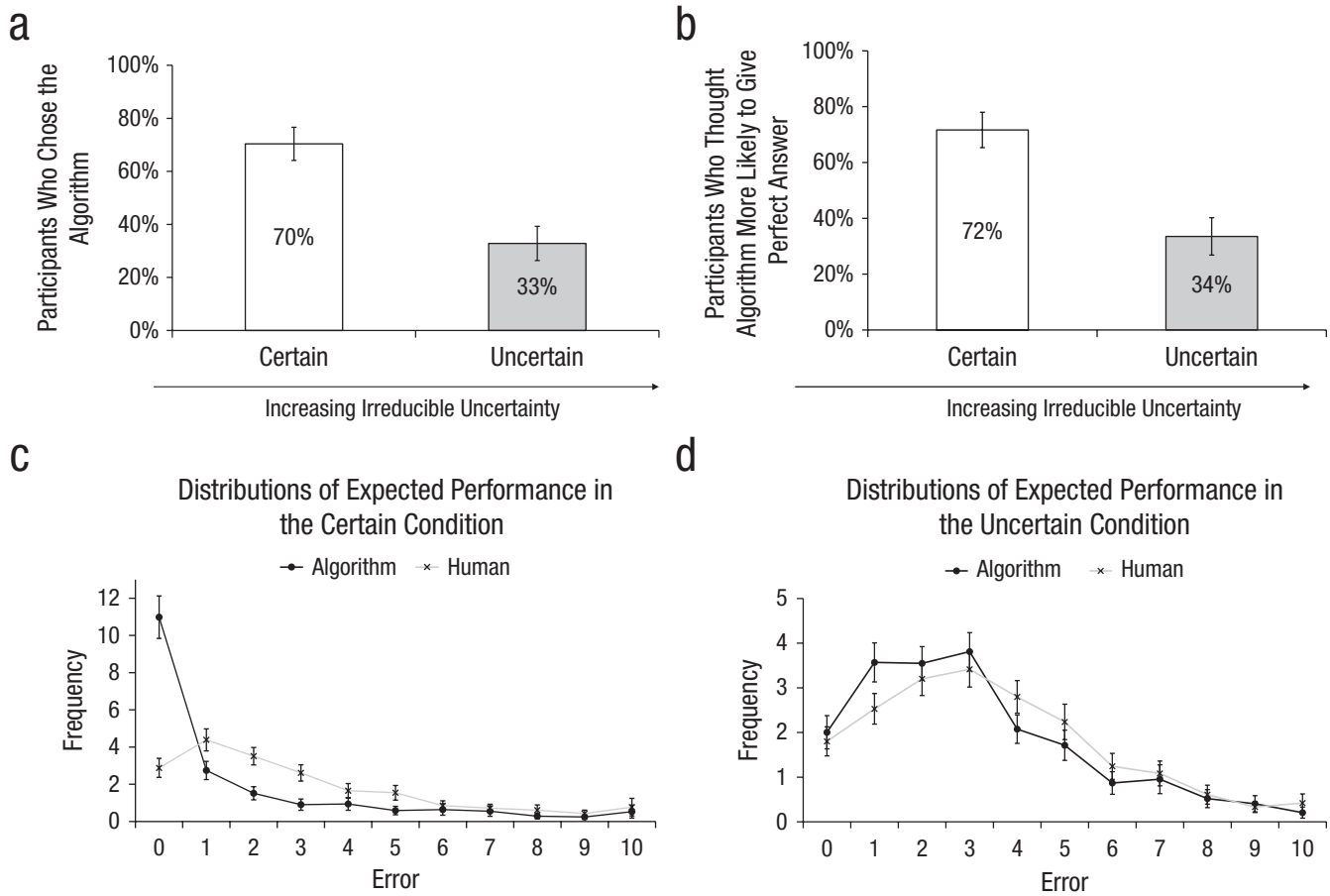


Fig. 5. Results for Study 4. The top row shows the percentage of participants who (a) chose to use the algorithm's answer and (b) thought the algorithm was more likely to give the perfect answer, separately for the certain and uncertain conditions. The bottom row shows the average distributions of participants' expectations for the model's performance and their own performance in 20 future forecasts, separately for the (c) certain condition and (d) uncertain condition. Error bars represent 95% confidence intervals.

$d = 1.80$; participants recognized that the model performed better on average, certain: $t(193) = -8.23$, $p < .001$, $d = -0.64$, uncertain: $t(190) = -4.17$, $p < .001$, $d = -0.26$; and participants reported more confidence in the model's answers on average, certain: $t(205) = 13.36$, $p < .001$, $d = 1.37$, uncertain: $t(203) = 3.63$, $p < .001$, $d = 0.26$, in both conditions. However, participants were substantially less likely to choose the model in the uncertain condition (33%) than the certain condition (70%), $\chi^2(1, N = 410) = 57.86$, $p < .001$, Cramer's $\phi = .38$ (see Fig. 5). This negative effect of uncertainty on participants' choices persisted in a logistic regression controlling for participants' estimates of their own and the model's average performance, $z(385) = 5.59$, $p < .001$, $OR = 3.78$, 95% $CI = [2.37, 6.03]$. As in Study 1, these results suggest that people may be unwilling to accept the best possible algorithm in inherently uncertain domains, even when they report that it offers better performance than human judgment on average.

Participants' distributions of performance beliefs, crafted from their 20 estimates of the model's future

performance and their own, revealed that their choice of decision-maker relates to their beliefs about perfect answers. We created a variable representing which method each participant believed to be more likely to provide a perfect answer. Seventy-two percent of participants thought that the model had a better chance of being perfect in the certain condition, but only 34% thought that the model was more likely to be perfect in the uncertain condition, $\chi^2(1, N = 385) = 56.17$, $p < .001$, Cramer's $\phi = .38$. Further, this variable mediated the negative effect of irreducible uncertainty on participants' choice of the model in a logistic-regression-based mediation analysis, 95% $CI = [0.042, 0.135]$.³ In other words, participants' decreased likelihood of choosing the model in the uncertain condition related to their decreased likelihood of believing that the model was more likely to give a perfect answer.

A follow-up analysis using two continuous mediators suggests that people give more weight to their beliefs about the likelihood of perfect forecasts than to their beliefs about average performance when selecting a

forecasting method. We set the continuous mediator representing perfect answers to the number of perfect forecasts participants expected the model to make minus the number of perfect forecasts that participants expected to make themselves. We set the continuous mediator representing average performance to the average error that participants expected from the model minus the average error that participants expected from themselves. In this logistic-regression-based mediation analysis, participants' beliefs about the method's perfect forecasts mediated the effect of their assigned condition on their choice of the model, 95% CI = [0.122, 0.314]; their beliefs about average performance did not, 95% CI = [-0.010, 0.078]. Consistent with diminishing sensitivity to forecasting error, the results of this analysis suggest that participants selected their forecasting method on the basis of the frequency with which they expected each method to make a near-perfect forecast, instead of on their expectations about average performance.

Studies 5a and 5b

Studies 3 and 4 showed that people are less likely to use algorithms in inherently uncertain domains because of diminishing sensitivity to error. One implication of this proposal is that people should be more likely to use an algorithm instead of human judgment when it offers higher variance in performance. We tested this prediction in Studies 5a and 5b.

Method

We preregistered that we would recruit 400 participants from Amazon Mechanical Turk (200 per condition) in both studies. In Study 5a, 469 participants clicked on the study link; 48 participants failed an attention check and were not allowed to begin the study, and 401 participants completed all dependent measures. In Study 5b, 527 participants clicked on the study link; 81 participants failed an attention check and were not allowed to begin the study, and 399 participants completed all dependent measures. The sample was 49% male and 37 years old on average in Study 5a and 52% male and 36 years old on average in Study 5b.

Study 5a. The task in Study 5a was the same as in Study 4; however, we manipulated the variance of the irreducible uncertainty in the task while holding the model's average performance constant (similar to Study 2). Participants were assigned to two conditions, *high variance* and *low variance*. The algorithm was always off by 1 in the low-variance condition and equally likely to be off by 0 or 2 in the high-variance condition. Both algorithms offered the same average performance and the same

earnings in expectation, but the algorithm in the high-variance condition offered higher variance in both performance and earnings.

Study 5b. Study 5b was similar to Study 5a, but we used the forecasting task from Study 3, which has a wider interval of outcomes (25–75), so we could offer participants a high-variance algorithm that never provided a perfect answer but clearly outperformed human judgment. In the low-variance condition, the algorithm was equally likely to be off by 6 or 7. In the high-variance condition, the algorithm was equally likely to be off by 2 or 11. We also changed the incentive scheme: Participants received \$0.60 if their incentivized prediction was off by 1 or less, and this bonus decreased by \$0.06 for each 2 units of error in their incentivized prediction.

In both studies, both algorithms offered the same average performance and earnings in expectation, but the algorithm in the high-variance condition offered higher variance in performance. As a result, if people have diminishing sensitivity to forecasting error, we would expect them to be more likely to choose the algorithm in the high-variance condition. Additionally, because the algorithm in the high-variance condition of Study 5b never made a perfect forecast or earned the best possible bonus, a positive finding in Study 5b would suggest that people's selection of forecasting methods is not completely driven by perfect performances.

Results

Study 5a. In Study 5a, the model performed significantly better than participants on average, high variance: $t(201) = 29.39, p < .001, d = 2.86$, low variance: $t(199) = 16.78, p < .001, d = 1.68$, and participants recognized that the model performed better on average, high variance: $t(201) = -8.43, p < .001, d = -0.55$, low variance: $t(199) = -4.10, p < .001, d = -0.25$, in both conditions of Study 5a. However, although participants had significantly greater confidence in the model's answers on average in the high-variance condition, $t(201) = 8.97, p < .001, d = 0.65$, they did not in the low-variance condition, $t(199) = 0.48, p = .634, d = 0.04$. Similarly, participants were significantly more likely to choose to use the model's answer in the high-variance condition (56%) than in the low-variance condition (26%; see Fig. 6), $\chi^2(1, N = 402) = 38.56, p < .001$, Cramer's $\phi = .31$. This positive effect of variance on participants' choice of the model persisted in a logistic regression controlling for participants' estimates of their own and the model's average performance, $z(402) = 5.17, p < .001, OR = 3.36, 95\% CI = [2.12, 5.32]$. Participants' relative preference for the higher-variance algorithm further supports the notion that they chose between forecasting methods on the basis of their expectations

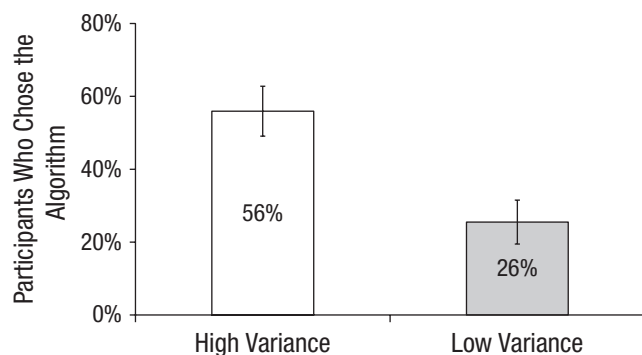


Fig. 6. Results for Study 5a: percentage of participants who chose the algorithm in the high-variance condition (algorithm off by 0 or 2) and low-variance condition (algorithm off by 1). Error bars represent 95% confidence intervals.

about the best possible performances, not on the basis of the average performance.

Study 5b. In Study 5b, the model performed significantly better than participants on average, high variance: $t(196) = 41.37, p < .001, d = 4.15$, low variance: $t(203) = 36.96, p < .001, d = 3.66$; participants recognized that the model performed better on average, high variance: $t(196) = -11.54, p < .001, d = -0.49$, low variance: $t(201) = -6.50, p < .001, d = -0.44$; and participants reported more confidence in the model, high variance: $t(196) = 14.29, p < .001, d = 1.14$, low variance: $t(201) = 7.65, p < .001, d = 0.59$, in both conditions of Study 5b. However, participants were significantly more likely to choose to use the model's answer in the high-variance condition (67%) than in the low-variance condition (49%; see Fig. 7), $\chi^2(1, N = 400) = 13.63, p < .001$, Cramer's $\phi = .18$. This positive effect of variance on participants' choice of the model persisted in a logistic regression controlling for participants' estimates of their own and the model's average performance, $z(399) = 3.49, p < .001, OR = 2.12, 95\% CI = [1.39, 3.23]$. As in Study 5a, these results are

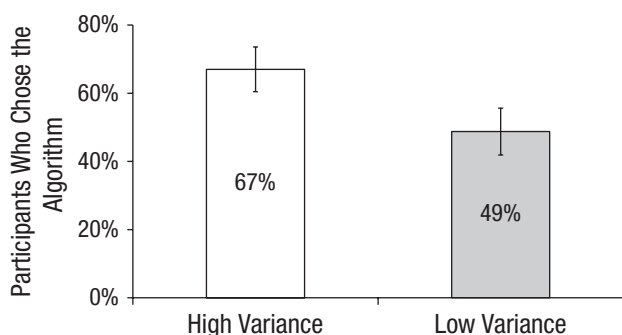


Fig. 7. Results for Study 5b: percentage of participants who chose the algorithm in the high-variance condition (algorithm off by 2 or 11) and low-variance condition (algorithm off by 6 or 7). Error bars represent 95% confidence intervals.

consistent with the notion that people choose between forecasting methods with diminishing sensitivity to forecasting error. Further, these results suggest that people's choices between forecasting methods are not driven solely by perfect performance; instead, these results (as well as the results of Study 1) are consistent with the notion that people have decreasing sensitivity to each marginal unit of forecasting error.

Discussion

Studies 1 through 5 provide convergent evidence that people have diminishing sensitivity to forecasting error and that this loss function leads them to prefer forecasting methods that they believe have a greater chance of generating a near-perfect forecast in uncertain domains (even at the expense of average performance). These risk-seeking preferences regarding error can lead people to underuse the best possible algorithm in favor of more error-prone options, such as human judgment. Thus, it appears that, instead of reflecting a preference for human judgment per se, people's apparent preference for human decision-makers in uncertain domains could reflect a preference for the higher expected variance of human performance.

These results suggest that convincing people to use algorithms in inherently uncertain domains is not a case of waiting until technology improves and algorithms perform better than they do today—people even underuse algorithms that offer the best performance possible. To the extent that driving, making medical decisions, predicting political events, and other types of decision-making are inherently uncertain, people may be unwilling to use even the best possible algorithms to make those decisions. The impact of this refusal is substantial, as society will not fully benefit from technological progress in consequential but uncertain domains until people are willing to use algorithms to make inherently uncertain predictions.

Transparency

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Author Contributions

Both authors contributed to the conceptual development of the work. S. Bharti created the study materials. B. J. Dietvorst analyzed the data, wrote the manuscript, and created the supplemental materials posted on OSF. S. Bharti reviewed and edited the manuscript and supplemental materials. Both authors approved the final manuscript for submission.

Declaration of Conflicting Interests

The author(s) declared that there were no conflicts of interest with respect to the authorship or the publication of this article.

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Open Practices

All data, analysis code, and materials have been made publicly available via OSF and can be accessed at <https://osf.io/erjq9/>. The design and analysis plans for the studies were preregistered on AsPredicted, and copies of the pre-registrations can be found at <https://osf.io/erjq9/>. The complete Open Practices Disclosure for this article can be found at <http://journals.sagepub.com/doi/suppl/10.1177/0956797620948841>. This article has received the badges for Open Data, Open Materials, and Preregistration. More information about the Open Practices badges can be found at <http://www.psychologicalscience.org/publications/badges>.



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Notes

1. We use the term *algorithm* to describe any tool that uses a fixed step-by-step decision-making process, including statistical models, actuarial tables, and calculators.
2. This article addresses prediction tasks, not classification tasks.
3. This difference persisted when this variable was coded as the percentage of participants who thought they were more likely to be perfect than the model. Twenty-five percent of participants thought that they had a better chance of being perfect in the uncertain condition, but only 10% thought that they were more likely to be perfect in the certain condition, $\chi^2(1, N = 385) = 15.75, p < .001$, Cramer's $\phi = .20$. Further, this variable mediated the negative effect of irreducible uncertainty on participants' choice of the model in a logistic-regression-based mediation analysis (95% CI = [0.027, 0.118]).

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